NESSIE VIETNAM

SCHALLENGE 2025

Trần Lê Ngọc Tiến

Trần Mỹ Duyên

Lê Thị Nguyệt

Trần Phú Anh Tuấn

FOURTUNES

EXECUTIVE SUMMARY

NESSIE

SITUATION

Nessie, a **leading global F&B group**, renowned for its **instant coffee** and **sustainability commitment**

Creating Shared Value strategy

NET ZERO **Roadmap by 2050**

COMPLICATION

Vietnam's Robusta output + **Nessie's single source dependency from Vietnam** → **Supply risks**

Suez Canal → rerouting via **Cape of Good Hope**

transit time **2–3** weeks

prone to damage during production and transport.

comply EU market regulations

QUESTION

How can Nessie **stabilize its supply chain and production costs**, while also **advancing CO_2 emission reduction** and **ensuring consistent product quality** to meet both domestic and EU market standards?

ANSWER

Diversifying and **Classifying** Robusta supply
+
Long-term supply contract

Protecting goods during transportation
+
Vision Inspection Technology
+
Finding an **alternative sourcing** option

Hire a **green 3PL** to stabilize domestic and international transport
+
Build a Flexible, Resilient & Green **Distribution Network**

ANALYSIS The current situation of the industry and the company

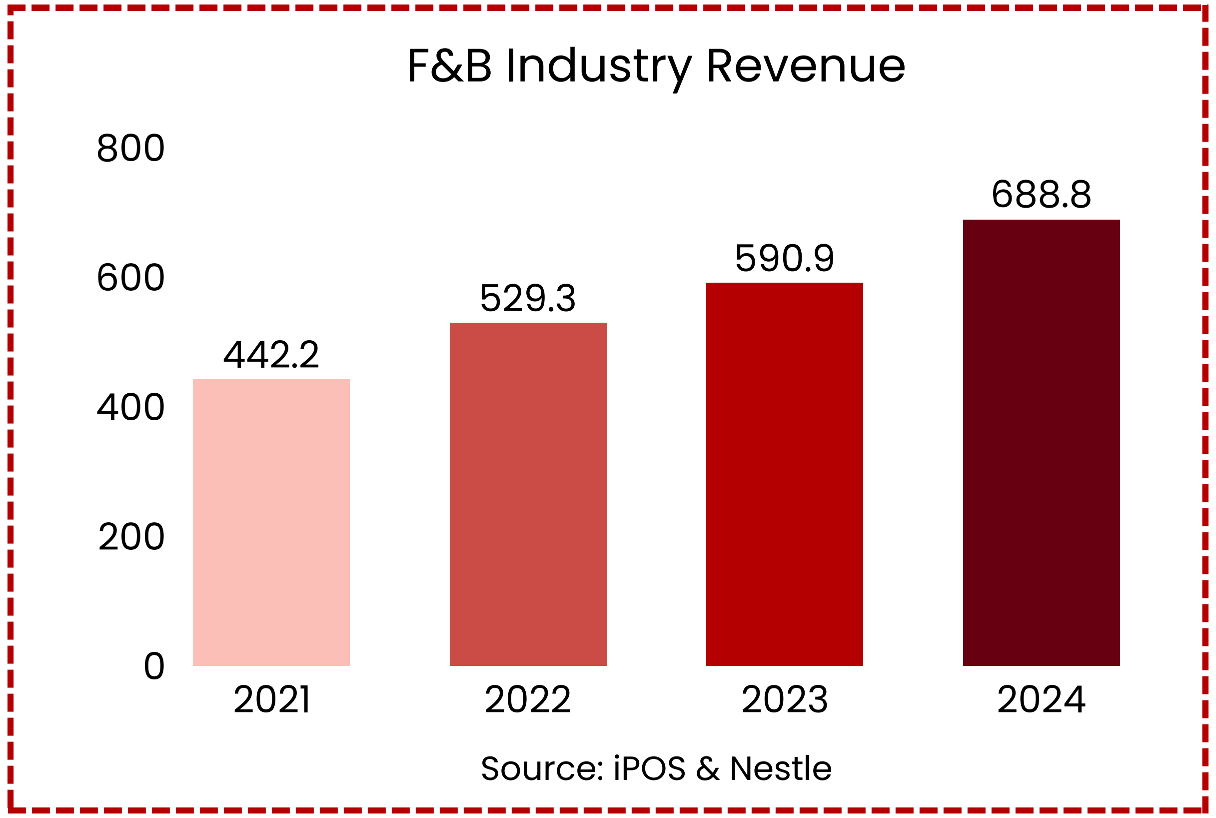
NESSIE

Market Analysis

Vietnam has the **highest F&B outlet** density in Southeast Asia, with about **323,010** stores. Around **32%** of consumers prefer dining at F&B chains whenever the need arises, and Gen Z accounts for **58%** of this group.

CAGR (2024-2028) **9.2%**

REVENUE 2024 **28.1** billion USD



NESSIE COMPANY PROFILE

NET ZERO 2050

Company

Nessie is one of the **world's largest food & beverage groups.**

Objective #1

Stabilize and diversify **supply sources**

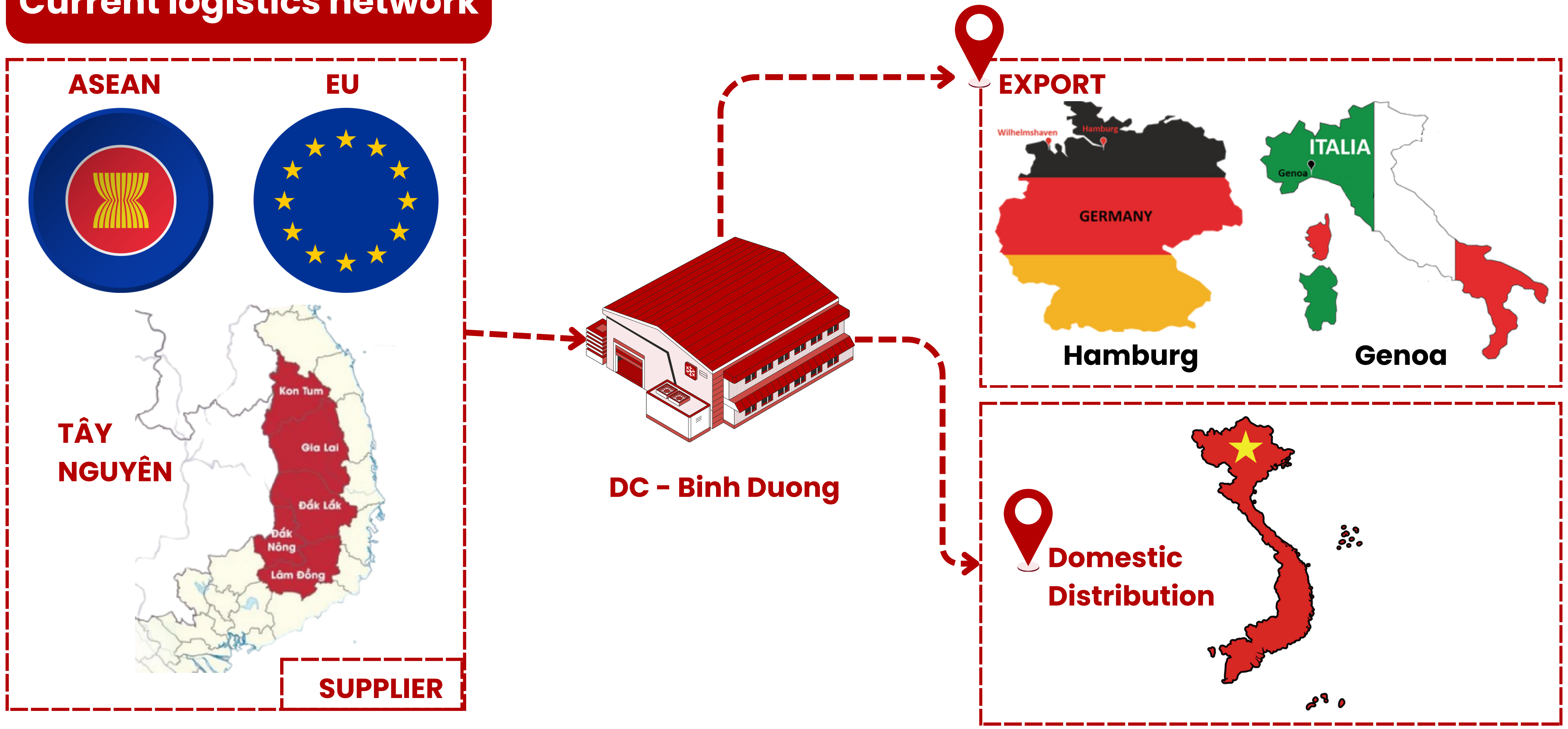
Objective #2

Reduce **CO_2 emissions**

Objective #3

Become a **production** and **export hub supplying coffee** to markets across Asia and Europe.

Current logistics network

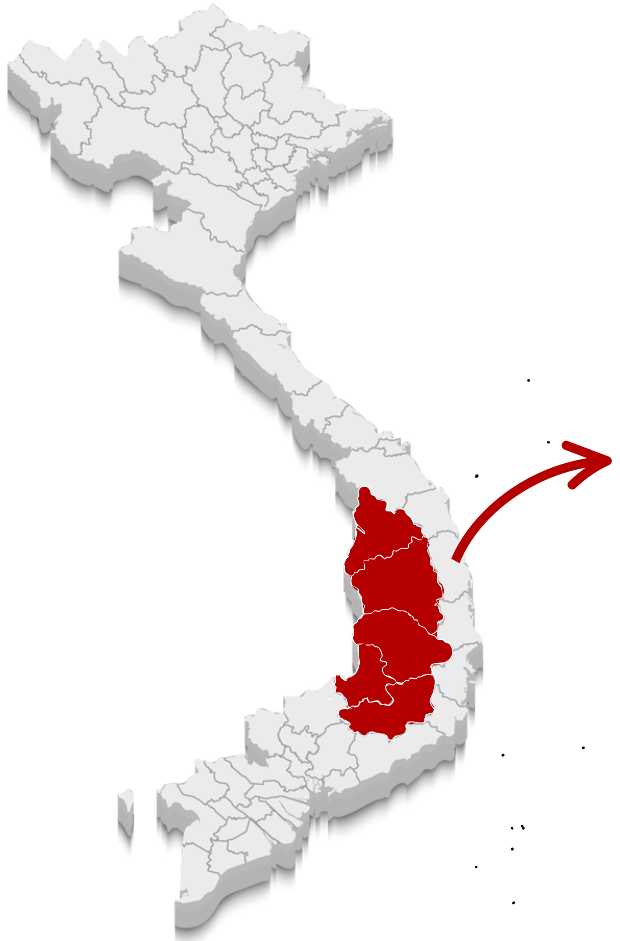


CURRENT ROBUSTA SUPPLY - CHALLENGES

Robusta green bean prices and output always fluctuate but Nessie just depends on a single supplier

NESSIE

Nessie's Robusta Source



Just Vietnam's Central Highlands

90% of Vietnam's total Robusta output

Climate change

Vietnam's coffee farmland

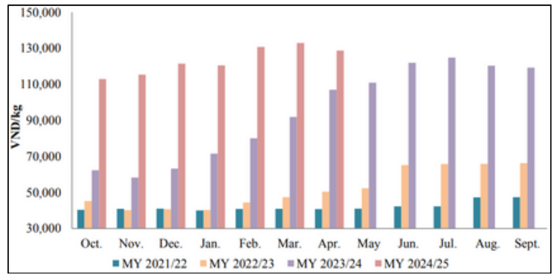
~50% by 2050

Global Robusta shortage

~35 million (60 kg) bags by 2040

Production costs

Vietnam's Coffee Prices



World's Coffee Prices

Vietnam's coffee farmland

65% year-over-year

Nessie's Production cost

~29% (VND/kg instant coffee)

High risk of severe supply disruption + **Potential raw material shortage in the future** + **Rising and volatile production costs**

Nessie benefits from its factory and DC being located in a major Robusta-growing region, optimizing logistics and ensuring quality. However, its **reliance on a single sourcing model makes the supply chain highly vulnerable to climate and price fluctuations**.

NESSIE'S PACKAGING - CHALLENGES

NESSIE

Manufacturer

~14,000 packaging enterprises in Vietnam

only **35%** focusing on green packaging

No qualified "green" packaging plants + Costs are 20- 30% higher = Unable to produce eco-friendly packaging at scale

Government

Vietnam's Environment Protection Regulations 2020 on EPR policy

Decree No.08/2022/NĐ-CP

Green transition program (MONRE, 2023)

Legal framework & support programs

Cost and **technology** barriers limit the implementation of green packaging production in Vietnam

Businesses in Vietnam are **unable** to mass-produce environmentally friendly packaging

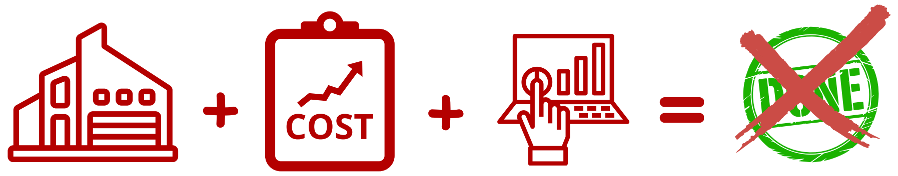
Nessie uses the packaging that be exported in **EU / ASEAN** and be easily damaged during production and transport

2% average damage rate for traditional packaging (plastic) → switching to recycled or compostable packaging

Scenario	Increase vs. Traditional	Estimated damage rate	Defected (per 10,000 packs)
Conservative	+30%	2.6%	260
Moderate	+75%	3.5%	350
Severe	+150%	5.0%	500

Using recycled or compostable packaging vs using traditional packaging (Plastic)

Packaging Component	Plastic Packaging	Recycled/ Compostable	Estimated Change
Laminate (Primary layer) - direct contact	2% damage	2.6 - 5% damage	**+ 30 - 150%**
Poly bag (Secondary layer) - grouping only	~ 0.5% damage	~ 0.5% damage	Insignificant



Nessie's ultra-premium instant coffee uses **recyclable packaging**, this packaging is **easily damaged** during production and transport. In addition, **high costs**, **dependence on imported materials**, and **limited local production capacity** create major challenges.

NESSIE'S TRANSPORTATION – CHALLENGES

NESSIE

Domestic Transportation

Factory & Distribution Center (Binh Duong) → Cat Lat port (Ho Chi Minh City)

100% road transport dependency

+ Bình Duong is **2ND** highest container traffic in VN

→ Frequent congestion + higher fuel use

+

International Route

Rerouted via Cape of Good Hope instead of Suez

~9,200 → ~12,500 nautical miles **(+36%)**

Transit time ↑ from **38 → 52 days**

+≈40% CO_2 per container

Risk from Yemen conflict expected to persist 1–2 years

→ ↑ Cost, ↑ Lead time, ↓ Carbon efficiency

+

Last-mile Distribution

Only **1 DC in Vietnam** → fully road-dependent

Only **2 DCs in Europe** → limited stock & slow response

Current capacity covers only **~75%** of potential demand

Current delay risk: **+3–5 days** per disruption

→ ↓ Service flexibility ↑ Lead time volatility

To sustain **+5%** YoY demand growth → need 2 new warehouse in EU

Estimated Cost Impact per FEU

Cost Category	Before Crisis (via Suez Canal)	After Crisis (via Cape of Good Hope)	Increase
Ocean Freight	$2,500	$3,800	52%
Bunker Adjustment Factor (BAF)**	$500	$700	40%
Inventory Carrying Cost[1]	$835 (for 38 days)	$1,140 (for 52 days)	36%
Total Estimated Cost / FEU	$3,835	$5,640	47%

[1]The inventory carrying cost is calculated based on an average cargo value of $50,000 per container and an annual holding cost rate of 15% of the cargo value. This cost increases because the goods remain at sea for 14 additional days.

Impact of Traffic Congestion on Domestic Transport

Indicator	Ideal Condition (No Congestion)	Actual Condition (Frequent Congestion)	Negative Impact
Distance	45 km	45 km	No change
Transit Time	1.5 hours	2.5 – 3 hours	+67% to +100%
Fuel Cost / Trip[3]	450,000 VND	600,000 VND	33%
CO_2 Emissions / Trip	95 kg CO_2	125 kg CO_2	31%

[3] Based on an average container truck fuel consumption of 25 liters/100 km and a diesel price of 24,000 VND/liter. Under congestion, fuel consumption rises due to frequent stop-and-go movement.

KEY DRIVER

The lack of **diversity and flexibility** in Nessie's logistics network makes the company **highly vulnerable** to supply chain disruptions, leading to **higher costs** and **increased CO_2 emissions**, which in turn undermines its ability to meet **sustainability commitments.**

CURRENT ROBUSTA SUPPLY – SOLUTIONS

SHORT TERM: Diversifying supply base

Understand the spend scope, cost drivers, and supply structure to set the basis for strategic sourcing decisions → Classify the importance of suppliers' products and decide sourcing strategy

NESSIE

DEFINE THE SPEND CATEGORY

Category Specifications

Main item: Green Robusta beans - Grade 1-2, **moisture ≤ 12.5%**

Other Components: Sugar, Milk powder, ...

End Customers and Markets

Domestic market: **Vietnam**

Export markets: **Hamburg** and **Genoa**

Logistics and Supply Chain Network

Supplier: Tây Nguyên → **Factory & DC**: Bình Dương → **DC**: Hamburg and Genoa → **Customer**

Transport modes:

- **Domestic: Intermodal** (Road transport >70%)
- **Export: Sea freight** from Cat Lai Port to Europe

Spend Visibility

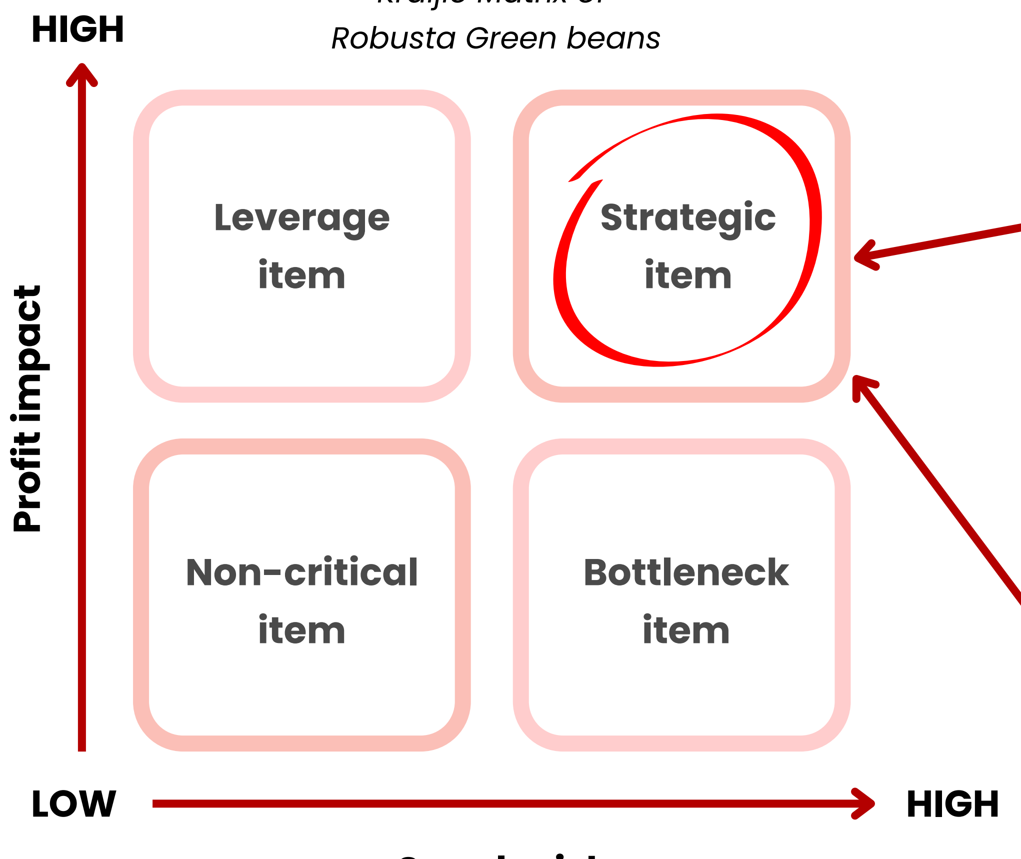
37.9% of total cost comes from coffee beans – the largest share in instant coffee production

65% year-on-year rise in Robusta prices makes procurement visibility and cost control critical

High Robusta bean cost share, export logistics complexity and **diverse markets** drive the need for **transparent spend visibility** and **sourcing clarity**

SOURCING STRATEGY DECISION

Kraljic Matrix of Robusta Green beans



HIGH PROFIT IMPACT

- Has a significant impact on the final product cost
- Defines the distinct flavor that influences customer satisfaction
- Directly affects the profit margin

HIGH SUPPLY RISK

- Dependence on a single region
- Strong impact from climate change
- High price volatility and supply risk

SOURCING STRATEGY DECISION

Foster **long-term partnerships**, pursue **joint innovation** and **value creation** and **integrate suppliers** into **strategic planning**

CURRENT ROBUSTA SUPPLY – SOLUTIONS

NESSIE

SHORT TERM: Diversifying supply base

Using AHP, Nessie identified strategic suppliers and defined their respective roles

AHP METHODOLOGY FOR CHOOSING AND CLASSIFIED SUPPLIERS

OJECTIVES

Identify **supply sources** for expansion and classify **strategic roles** to address the **shortage** and 65% increase in domestic **raw material prices** compared to the previous year, as well as **future volatility.**

CRRITERIA

- **1.** Geographic Proximity & Logistics Advantage
- **2.** Supply Capability & Stability
- **3.** Traceability System Transparency
- **4.** Quality Consistency among Domestic Suppliers
- **5.** Sustainability Certification Coverage
- **6.** EVFTA & Regional Trade Benefits
- **7.** Political & Economic Stability

OPTIONS

Laos	Indonesia	Thailand
Malaysia	India	Brazil
Uganda	Côte d'Ivoire	Tanzania

WEIGHTING

1	2	3	4	5	6	7
25%	20%	15%	15%	10%	10%	5%

AHP Scoring

Strategic Supplier Classification

Strategic Role	Country	% of Total Production	Management Approach
Core Strategic Supply	Vietnam, Indonesia, Thailand	60-70%	Long-term partnership. Core supply ensuring quality and EU traceability
Flexible Supply	Brazil, India, Malaysia, Laos	20-25%	Flexible sourcing to respond to market fluctuations
Development Supply	Uganda, Côte d'Ivoire, Tanzania	5-10%	Pilot supply, potential for future expansion

Structured & climate-resilient sourcing diversification to secure supply continuity and cost stability

Expected Outcome

- **Delivery delays ~ 30%**
- **Price Volatility 28% ▶ 18%**
- **Cost risk**
  If green coffee **prices rise by 65%** YoY and Long term supply **contract coverage** at:
  - 50% volume → ~32.5%
  - 70% volume → ~45%
  - 100% volume → 65%

LONG TERM

Long-Term Supply Contracts

- Ensure **continuous supply** and avoid **raw material shortages**
- **Stabilize prices** and **optimize procurement costs**
- **Strengthen supplier relationships** to enhance **supply chain efficiency**

Long-Term Supply Contracts are crucial for Nessie's strategy of diversifying its supply base

NESSIE'S PACKAGING – SOLUTIONS

NESSIE

SHORT - TERM

Damage Prevention & Quality Checks for Manufactured Products

1. Apply **Vision Inspection** technology for Quality Check

2. Add **protective cushioning** (e.g., bubble wrap or pallet foam) inside containers

3. Label all export cartons with **"Fragile / Keep Dry"** warnings

Protect goods during transportation, **minimize the risk of damage**, and ensure products reach customers intact and in high quality.

LONG - TERM

By 2030, all imported packaging to the EU must be **100%** recyclable or achieve **≥70%** actual recycling rate.

⇨ **NESSIE** needs to:

Objective #1
Shift to green & recyclable materials packaging

Objective #2
Ensure premium quality & product protection

Objective #3
Align with EU environmental standards and brand sustainability goals

Materials

Film Structure	Main Components	Recyclability	Durability / Shelf Life
BOPP/PE	BOPP (outer) + PE (inner)	★★★★ (Mono-polyolefin ⇒ easy to recycle)	★★★
PET/PE	PET (outer) + PE (inner)	★★★ (Different polymers ⇒ less recyclable)	★★★★
BOPP/PP	Two layers of PP-based films	★★★★★ (100% mono-material)	★★★★
NY/PE (PA/PE)	Nylon + PE	★★	★★★★★
BOPP(PET)/PE/AL/PE	BOPP + PET + Aluminum + PE	★	★★★★★
Paper/PE/AL/PE	Paper + Aluminum + PE	★	★★
PET/PE/AL/PE/LLDPE	PET + Aluminum + multi-PE layers	★	★★★★★

Select alternative materials to replace the old laminate layers

Based on technical performance and recyclability ranking, **two-layer packaging structures (BOPP/PE or BOPP/PP)** are the most suitable options - offering an optimal balance between durability, recyclability, and production cost.

NESSIE'S PACKAGING - SOLUTIONS

NESSIE

LONG - TERM

ATIGA Tariff 0% : ASEAN-origin goods (incl. packaging) enjoy 0% import tax + logistics cost→ **lower cost vs. EU sourcing**.

~64% of coffee producers already use sustainable packaging

Prioritize green packaging manufacturers within the ASEAN region.

Country	Key Strength	Example Supplier	Advantage
Thailand	Leading regional packaging hub	SCG Packaging (SCGP)	Advanced tech (mono-material, bioplastic), near Vietnam → lower logistics & CO_2
Malaysia	Bio-fiber / palm-based bioplastic	Ean Label	Emerging green tech, lower logistics cost
Vietnam	Local cost advantage	Tetra Pak	Easy coordination, partial green capability

STRATEGIC ACTIONS

1. Select sustainable partners (SCGP, Tetra Pak VN, Ean Label).
2. Pilot green packaging (mono-material or bio-coated paper).
3. Negotiate long-term partnerships with CO_2 reduction commitments.
4. Upgrade packaging lines to meet EU PPWR 2030 standards.

EXPECTED OUTCOME

Indicator	Before	After	Δ% / Impact
Green packaging share	15%	65–70%	↑ +50 pts
CO_2 emissions	3,000 kg/t	1,600 kg/t	↓ 45%
EU compliance rate	60%	95%	↑ +35 pts
Customs clearance	100%	85–90%	↓ 10–15%
Brand green image	40%	90%	↑ +50 pts
Export value	–	+10–15%	↑ due to premium perception

EFE MATRIX

Criteria Group	Description	Weight
1. Environmental Standards & Green Certifications	ISO 14001, FSC, recycling, carbon neutral	20%
2. Technology & Innovation Capability	Mono-material, bio-based film, PLA-coated	20%
3. Supply Stability & Production Capacity	Scale, output, on-time order fulfillment	15%
4. Cost & Logistics Advantage	Unit cost, proximity to Vietnam (tax, shipping)	15%
5. Brand Reputation & International Partnerships	Long-term FMCG partners, credit rating	10%
6. Co-Innovation Capability	Willingness for joint R&D, tech sharing	10%
7. EU/PPWR Export Compliance	EPR, Circular Design, ≥90% recyclability	10%
Total		**1**

10–40 Scale: 100 = Weak | 200 = Average | 300 = Fair | 400 = Good

External Factors	Weight	SCG Packaging	Tetra Pak	Ean Label
Environmental Standards & Green Certifications	20%	40	40	30
Technology & Innovation Capability	20%	40	35	30
Supply Stability & Production Capacity	15%	40	40	25
Cost & Logistics Advantage	15%	35	30	40
Brand Reputation & International Partnerships	15%	40	40	25
Co-Innovation Capability	15%	35	40	25
EU/PPWR Export Compliance	15%	40	40	25
Weighted Score	1	**385**	**375**	**295**

SCG Packaging (Thailand) is the preferred alternative partner for green, environmentally friendly packaging.

NESSIE'S TRANSPORTATION – SOLUTIONS

NESSIE

SHORT - TERM

Hire a green 3PL service provider to quickly stabilize both domestic and international transportation operations

GREEN 3PL OPTIONS

3PL Provider	Key Strengths	Main Locations	Green Highlights
DHL	EU Top 1; ISO 14001 certified; electric & biofuel fleet	Hamburg, Genoa, Milan	Transparent CO_2 reporting; investment in renewable energy
DB Schenker	Strong EU network; 100% green in German warehouses	Hamburg, Bremen	"Carbon Neutral 2040" commitment
Maersk	Integrated sea + warehousing + inland transport	Genoa, Rotterdam	Up to 80% CO_2 reduction using biofuel
Geopost	Strong in last-mile electric delivery	Hamburg, Paris, Milan	100% carbon-neutral delivery in 10 EU countries

WHY DHL?

- **Existing hubs in Hamburg & Genoa** (aligned with Nessie's two DCs)
- Certified ISO 14001, implements **GoGreen Plus** (biofuel program)
- Covers **ocean, warehousing**, and **last-mile** operations
- **42,000 electric vehicles** with transparent CO_2 reporting
- Trusted by **Nestlé, Unilever, and Danone**

EVALUATION

3PL Provider	Network	Green Compliance	Cost Stability	Strategic Fit	Total /20
DHL	5	5	4	5	**19 / 20**
DB Schenker	4	5	4	3	16 / 20
Maersk	3	5	3	4	15 / 20
Geopost	2	4	4	3	13 / 20

EXPECTED OUTCOME

Indicator	Before	After	Δ% / Impact
Emission Reduction	Baseline CO_2e	CO_2e ↓ 80–90% (ocean freight), ↓ 15–20% total order	Significant reduction in carbon footprint
Operational Efficiency	Current lead time & OTD	Lead time ↓ 30–35%, OTD ≥ 95%	Faster delivery, higher on-time performance
Cost Stability	Existing cost fluctuation	Logistics cost fluctuation ≤ ±10%/year	More predictable logistics cost
ESG & Compliance	Partial EU compliance	Fully compliant with EU Green Freight & CO_2 reporting	Meets EU sustainability & traceability requirements
Brand Impact	Low visibility / recognition	Book & Claim Certificate → "Green Supply Chain Partner"	Stronger ESG reputation & brand positioning

NESSIE'S TRANSPORTATION – SOLUTIONS

LONG – TERM

Build a Flexible, Resilient & Green Distribution Network

Reasons

De-risking & Enhancing Resilience

In Vietnam: Launch a **Northern hub** (Hai Phong) to **decongest and reduce** dependency on the overloaded Cat Lai area.
In Europe: Two **warehouses** in Hamburg and Genoe eliminate **single-route risks** (the Suez Canal), ensuring continuous product flow.

Enhancing Customer Service

Market Proximity: Holding inventory in Europe significantly shortens last-mile **delivery times.**
Model Shift: Move from a **reactive** (waiting for shipments from VN) to a **proactive** model (shipping directly from EU hubs) for an immediate response to market fluctuations.

Category	Before	After	Improvement / Δ%
Average Lead Time	45 days	~30 days	↓ 30–35%
On-Time Delivery	80%	≥95%	15%
Fill Rate	85%	≥95%	10%
CO_2 Emissions / Order	100%	80–85%	↓ 15–20%
Logistics Cost	±25% / year	±10% / year	↓ 15%

Phase

Phase 1: Feasibility & Planning

1. Conduct detailed analysis to select optimal warehouse locations in Hai Phong, Hamburg, and Genoa.
2. Partner Vetting: Select construction & 3PL partners
3. Financial Modeling: Develop detailed CAPEX (Capital Expenditure) and OPEX (Operating Expenditure) forecasts.

Phase 2: Construction & Setup

NORTHERN HUB (VN)
Construct or lease the facility and implement the Warehouse Management System (WMS)

EU HUBS
Finalize contracts, establish operational workflows, and integrate IT systems with European partners

Phase 3: Go-Live & Optimization

PHASED ROLLOUT
Pilot the Northern Hub and new logistics flows.

FULL OPERATION
Transition the entire network to the new operational model.

CONTINUOUS IMPROVEMENT
Ongoing KPI monitoring and optimization of cargo flows

IMPLEMENT PLAN

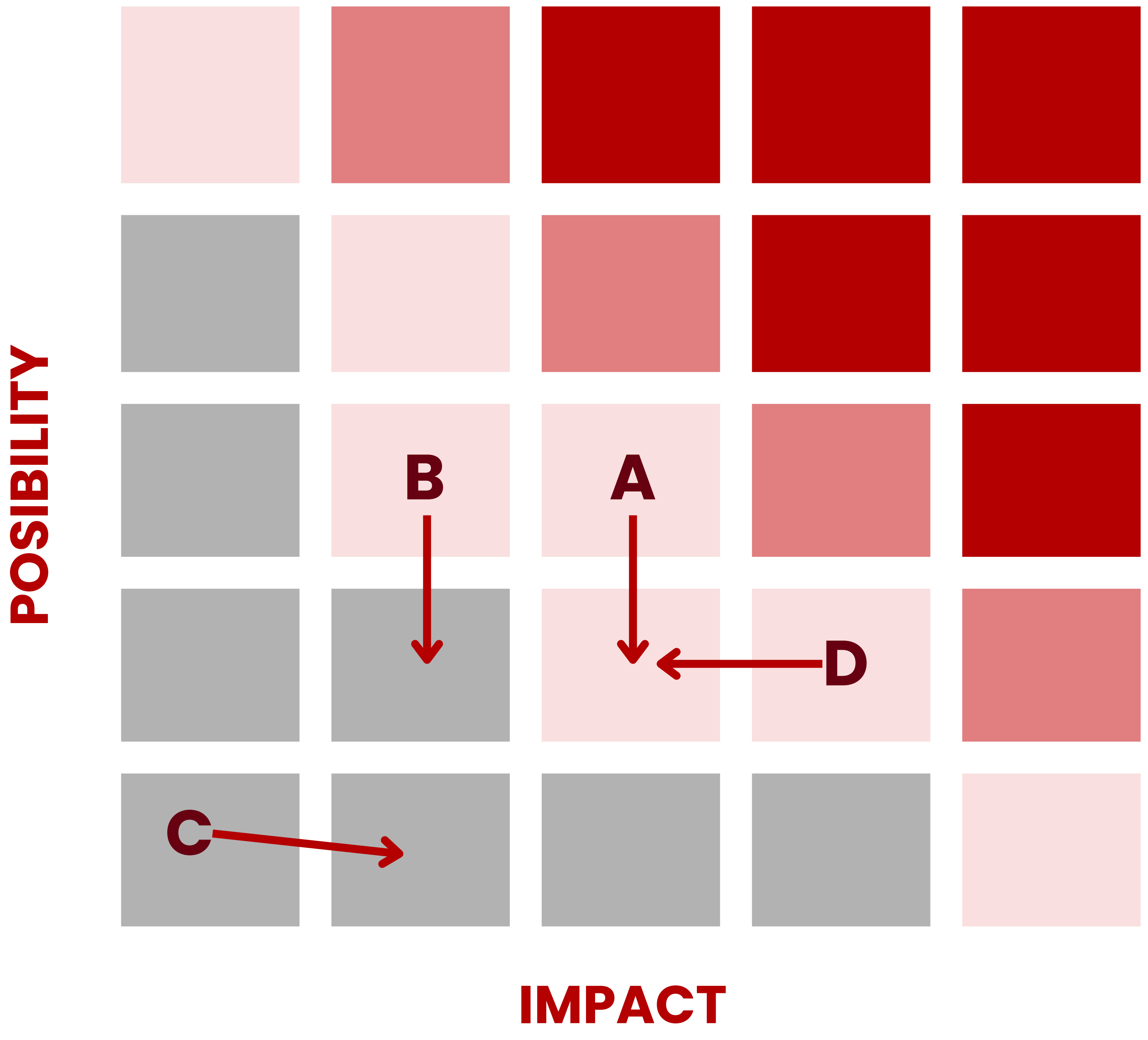
NESSIE

Supply Chain Capacity Enhancement Action Plan

	SHORT - TERM	LONG - TERM			
MILESTONE	2025 - 2026	2027	2028	2030	2050
STRATEGY	SUSTAINABLE PARTNERSHIP & SOURCING FOUNDATION	DRIVING EFFICIENCY TOWARD A GREEN SUPPLY CHAIN			
ACTIVITY	PLANNING AND SELECT POTENTIAL 3PLS	CLOSELY COLLABORATE WITH 3PLS, SET UP CLEAR SLA AND MANAGE PERFORMANCE FREQUENTLY		SET UP OWN WAREHOUSE FOR EASIER MANAGEMENT AND COST REDUCTION	
	EXPAND SUPPLIER NETWORK	LONG-TERM CONTRACTS & STRATEGIC PARTNERS			
	STRENGTHEN INSPECTION AND MONITORING	INTRODUCE NEW PACKAGING			
	ACTIVELY PURSUE R&D INITIATIVES AIMED AT INNOVATING ENERGY-SAVING AND EMISSION-REDUCING TECHNOLOGIES				
OUTCOME	Mitigating immediate **supply risks**; ↑ **Quality control**; Established a resilient **operational foundation**	Supply disruption risk ↓ ~ 30%; Total CO_2 emission ↓ ~ 7%		NET ZERO 2050	

OVERVIEW | PROBLEMS | SOLUTIONS | **IMPLEMENTATION** | RISK MANAGEMENT

RISK MANAGEMENT



Code	Main Risk	Mitigations
A	Delay or disruption from suppliers (Vietnam, ASEAN, EU)	• Expand the **supplier network** to reduce dependence on a single source. • Establish **long-term supply contracts** with strategic suppliers. • Maintain **safety stock** for key raw materials.
B	Inefficient warehousing and distribution (internal DC or 3PL hub)	• Implement **Warehouse Management System** (WMS) and quality inspections. • Strengthen collaboration with **3PLs**, define clear and regular **SLA performance reviews**. • **Long-term:** Establish a dedicated **Nessie warehouse** for better cost and quality control.
C	Product damage or loss during transportation (domestic & export)	• **Upgrade packaging** to ensure product protection during transit. • Apply **specialized handling procedures** for export shipments. • Monitor **logistics performance** and implement **corrective actions promptly.**
D	Variation in Robusta quality (moisture, flavor, size, density) across new sources — affecting blend consistency and product quality.	• Establish internal **"Quality Benchmark"** standards to ensure consistency. • Conduct **blend testing & pilot runs** to verify flavor and stability.


NESSIE
THANK YOU

REFERENCE

1. Hapag-Lloyd. (n.d.). Bunker Surcharge. Retrieved October 16, 2025, from https://www.hapag-lloyd.com/en/products/fleet/bunker-surcharge.html
2. Kenton, W. (2023, July 26). Carrying Costs: What They Are and How to Calculate Them. Investopedia. Retrieved October 16, 2025, from https://www.investopedia.com/terms/c/carrying-costs.asp
3. U.S. Environmental Protection Agency. (2024, May). Greenhouse Gases Equivalencies Calculator - Calculations and References. Retrieved October 16, 2025, from https://www.epa.gov/energy/greenhouse-gases-equivalencies-calculator-calculations-and-references
4. https://tapchikinhtetaichinh.vn/fb-viet-nam-thi-truong-tiem-nang-cho-cac-doanh-nghiep-nuoc-ngoai.html https://www.ewadirect.com/proceedings/aemps/article/view/14195?fbclid=IwY2xjawNbfyJleHRuA2FlbQIxMABicmlkETF0MUtQSG5EZnpPSTFHR3h3AR6K9gOBXFWyL0xkbYA3ZBdfgj0v_Bo9IC49JVFarlN1uPKW-zsDNPkZaGpkqg_aem_kqQ1sndGrfae8Er496q9bA https://asana.com/resources/risk-matrix-template
5. DB Schenker. (n.d.). Sustainability. Retrieved October 16, 2025, from https://www.dbschenker.com/global/meta/sustainability
6. DHL. (n.d.). GoGreen Plus - Making sustainable logistics happen. Retrieved October 16, 2025, from https://www.dhl.com/global-en/home/press/specials/gogreen-plus.html
7. Geopost. (n.d.). Our commitment to sustainability. Retrieved October 16, 2025, from https://www.geopost.com/sustainability/
8. Maersk. (n.d.). Decarbonisation in logistics. Retrieved October 16, 2025, from https://www.maersk.com/sustainability/decarbonisation
9. Chính phủ nước Cộng hòa xã hội chủ nghĩa Việt Nam. (2022). Nghị định số 08/2022/NĐ-CP của Chính phủ: Quy định chi tiết một số điều của Luật Bảo vệ môi trường.
10. McKinsey & Company. (2025). The Future of Sustainable Packaging in Southeast Asia.
11. Bộ Tài nguyên và Môi trường. (2023). Chương trình chuyển đổi xanh quốc gia.
12. Quốc hội nước Cộng hòa xã hội chủ nghĩa Việt Nam. (2020). Luật Bảo vệ Môi trường số 72/2020/QH14.
13. Hiệp hội Bao bì Việt Nam (VINPAS). (2025). Báo cáo ngành bao bì Việt Nam năm 2024.
14. ASEAN Secretariat. (n.d.). ASEAN Trade in Goods Agreement (ATIGA). Retrieved October 16, 2025, from https://investasean.asean.org/asean-trade-in-goods-agreement-atiga/
15. European Commission. (n.d.). Proposal for a new Regulation on packaging and packaging waste. Retrieved October 16, 2025, from https://environment.ec.europa.eu/publications/proposal-packaging-and-packaging-waste_en
16. ASEAN Secretariat. (n.d.). ASEAN Trade in Goods Agreement (ATIGA). Retrieved October 16, 2025, from https://investasean.asean.org/asean-trade-in-goods-agreement-atiga/
17. European Commission. (n.d.). Proposal for a new Regulation on packaging and packaging waste. Retrieved October 16, 2025, from https://environment.ec.europa.eu/publications/proposal-packaging-and-packaging-waste_en
18. Vietnam Coffee-Cocoa Association (VICOFA). (2025). 2024 Annual Report on Sustainable Practices in the Vietnamese Coffee Industry.
19. Chen, S. (2024, July 25). Analysis of the cost structure of Luckin Coffee. Guangdong University of Foreign Studies.
20. ReportLinker. (2023). Global Robusta Green Coffee Production by Country.
21. United States Department of Agriculture, Foreign Agricultural Service. (2025). Coffee: World markets and trade report.
22. Johnson, S. (2024, July 7). Vietnamese robusta bean production declining, hiking up price of instant coffee. ABC Asia.

APPENDIX

APPENDIX AHP Scoring method

Country	EVFTA & Regional Trade Benefits	Supply Capability & Stability	Geographic Proximity & Logistics	Sustainability Certification Coverage	Quality Consistency	Traceability Transparency	Political & Economic Stability	Total Weighted Score
Indonesia	5	4	3	3	3	3	3	0,1425
Brazil	1	5	1	4	4	4	4	0,1345
Thailand	5	1	4	3	3	3	4	0,1283
Laos	5	2	4	3	2	2	3	0,1221
Malaysia	5	1	4	2	2	3	4	0,1174
India	1	3	3	2	3	2	3	0,1069
Uganda	1	3	2	2	2	2	2	0,0886
Côte d'Ivoire	1	2	2	2	2	2	2	0,0799
Tanzania	1	2	2	2	2	2	2	0,0799

APPENDIX Expected Outcome

KPI (Outcome)	Before	After (Estimate)	% Change / Δ (Calculation)
Share of "Green" Packaging	15%	65–70% (avg. 67.5%)	Δ = (0.675-0.15)/0.15 = +350%
CO_2 Emissions (kg/ton material)	3000	1600	(3000-1600)/3000 = -46.7%
EU PPWR Compliance Rate	60%	95%	+35 percentage points (+58.3%)
Customs Clearance Time (EU)	100% baseline	85–90% (avg. 87.5%)	(100-87.5)/100 = -12.5%
Green Brand Perception Score	40%	90%	(0.9-0.4)/0.4 = +125% (+50 pp)
Export Value Growth (Estimated)	Baseline	+10–15%	(After-Before)/Before = +10–15%
Logistics Defect Rate	20%	1.5–2.6% (varies by scenario)	Example: 2.0→1.5% = -25%

APPENDIX: CURRENT ROBUSTA SUPPLY - CHALLENGES

Vietnam's coffee farmland

65% year-over-year

Nessie's Production cost

~29% (VND/kg instant coffee)

"On average, a 10-cent increase in the cost of a pound of green coffee beans in a given quarter results in a 2-cent increase in manufacturer and retail prices in the current quarter."

Leibtag, E. (2007). Cost pass-through in the US coffee industry (No. 38). USDA Economic Research Serv.

APPENDIX: EXPECTED OUTCOMES OF CURRENT ROBUSTA SUPPLY SOLUTIONS

Expected Outcome

of supplier diversification

Delivery delays ~ 30%

"In the event of supply disruptions, companies that diversified their supplier base experienced a 30% reduction in delivery delays."

Celestin, M., & Sujatha, S. (2024). Impact of global supply chain disruptions on business resilience: Strategies for adapting to pandemics and geopolitical conflicts. International Journal of Advanced Trends in Engineering and Technology, 9(2), 44-53

APPENDIX Logical Basis for Criteria Selection

Category	External Factors (Opportunities/Threats)	Scientific/Strategic Rationale
Sustainability compliance	Environmental regulations (EPR, FSC, ISO 14001)	Aligns with Environmental Management Systems (EMS) and sustainability reporting (ISO 14031, GRI
Technological capability	Advanced barrier & sealing technology	Ensures packaging integrity (referencing ASTM F88/F1921 seal standards).
Cost competitiveness	Total landed cost, localization level	Considers supply chain cost theory (Porter's value chain, 1985).
Supply reliability	Lead time, logistics stability	Based on SCOR Model (Plan-Source-Make-Deliver) performance indicators.
Innovation & R&D	Capacity to develop compostable/mono-material packaging	Follows Dynamic Capabilities Theory (Teece, 1997) - innovation as competitive advantage.
Partnership potential	Long-term contract flexibility, collaboration willingness	Supported by Strategic Alliance Theory and relational contracting literature.

KEY TERM

ATIGA (ASEAN Trade in Goods Agreement): This is a free trade agreement among ASEAN member states. Under ATIGA, qualified goods originating from and traded within ASEAN are eligible for a 0% import tariff. This provides a significant cost advantage over sourcing similar materials from the EU, which would be subject to standard import duties and higher logistics costs.

GLOSSARY OF KEY VIETNAMESE REGULATIONS

Regulation / Program	Description	Impact on Nessie
EPR Policy (in Env. Protection Law 2020)	Extended Producer Responsibility. This policy mandates that manufacturers and importers must be responsible for the entire lifecycle of their products, including the collection and recycling of packaging after consumption.	Forces Nessie to invest in or contribute to recycling schemes, making the choice of packaging material a critical financial and operational decision.
Decree No. 08/2022/ND-CP	This decree provides the detailed roadmap and specific recycling rates required for implementing the EPR policy.	Sets clear, legally binding targets that Nessie must meet, increasing the pressure to adopt easily recyclable materials.
Green Transition Program (MONRE, 2023)	A national program by the Ministry of Natural Resources and Environment to support businesses in adopting sustainable practices through financial incentives, technology transfer, and policy support.	Creates opportunities for Nessie to receive potential support (e.g., tax breaks, grants) if it invests in green packaging technology and infrastructure.

APPENDIX

TECHNICAL GLOSSARY OF PACKAGING MATERIALS

Abbreviation	Full Name	Description
BOPP	Biaxially Oriented Polypropylene	A durable plastic film with excellent moisture barrier properties. Widely used for food packaging.
PET	Polyethylene Terephthalate	A clear, strong plastic with good gas and moisture barrier properties.
PE	Polyethylene	The most common plastic, used as a sealing layer in laminates.
AL	Aluminum Foil	Provides the best barrier against light, oxygen, and moisture, ensuring maximum shelf life.

GLOSSARY OF TERMS

Term	Definition
FEU (Forty-foot Equivalent Unit)	A standard shipping unit equivalent to one 40-foot container, used to measure ocean freight volume.
BAF (Bunker Adjustment Factor)	A fuel surcharge applied by carriers to compensate for fluctuations in global oil prices.
Inventory Carrying Cost	The hidden cost of holding inventory, including capital cost, insurance, storage, and risk of obsolescence.
DC (Distribution Center)	A warehouse facility used for storing and redistributing goods to final destinations.

DETAILED CALCULATIONS

Description	Formula / Calculation	Result
Annual Inventory Carrying Cost	$50,000 (inventory value) × 15%	$7,500/year
Daily Carrying Cost	$7,500 ÷ 365	$20.55/day
Cost for 38-day Transit (via Suez)	$20.55 × 38	≈ $835
Cost for 52-day Transit (via Cape of Good Hope)	$20.55 × 52	≈ $1,140
Domestic Fuel Cost – Ideal	(45 km ÷ 100) × 25 L × 24,000 VND/L	≈ 450,000 VND/trip
Domestic Fuel Cost – Congestion	450,000 × 133%	≈ 600,000 VND/trip

ESTIMATED CO_2 EMISSION REDUCTION FORMULA

$$\mathrm{CO_2e\ Reduction\ (\%)} = \frac{\text{Baseline Emissions} - \text{Emissions using SMF}}{\text{Baseline Emissions}} \times 100$$

The estimated 80–90% CO_2e reduction is calculated based on Nessie’s commitment to use ocean freight services powered by Sustainable Marine Fuel (SMF) provided by DHL. This figure represents the maximum certified reduction under the GoGreen Plus program.

Source: Based on DHL’s GoGreen Plus program.

APPENDIX

NESSIE

SCORING RUBRIC

Criteria	5 Points (Excellent)	4 Points (Good)	3 Points (Fair)	2 Points (Basic)	1 Point (Unsatisfactory)
Network	Extensive EU-wide network, with available hubs in Germany & Italy, offering multi-modal services.	Broad EU network with a hub in either Germany or Italy.	Good network but lacks hubs in Nessie's strategic locations.	Limited network, strong only in a few countries.	Very weak network with no presence in key markets.
Green Compliance	ISO 14001 certified, provides transparent CO_2 reporting, and offers sustainable fuel solutions (e.g., GoGreen Plus).	Has a Net-Zero commitment and CO_2 reporting, but sustainable fuel solutions are limited.	Has green policies but lacks detailed CO_2 reporting for clients.	No clear green commitment or solutions.	Absolutely no environmental policies or commitments.
Cost Stability	Offers long-term contracts with a stable pricing mechanism, fluctuation below ±10%/year.	Offers long-term contracts but with higher fuel surcharge volatility.	Mainly offers volatile spot rates.	Unstable pricing.	No relevant experience, unable to integrate systems.
Strategic Fit	Proven F&B industry experience, easy IT integration, with major partners as references.	Experience with consumer goods but less F&B-specific experience.	IT system requires significant customization for integration.	No clear experience in the F&B industry.	No relevant experience, unable to integrate systems.

PRELIMINARY COST ESTIMATION (CAPEX & OPEX)

Category	Estimated Cost (Preliminary)	Notes
CAPEX (Initial Investment Cost)		
- Lease/Construction of Northern Hub (VN)	$2-3 million	Includes basic construction and land costs.
- Warehouse Management System (WMS)	$500,000	For all 3 locations.
OPEX (Annual Operating Cost)		
- Warehouse Leasing & Staffing Costs (EU)	$1.5-2 million / year	Outsourced 3PL warehouse operations in Hamburg & Genoa.
- Warehouse Operating Costs (VN)	$500,000 - $700,000 / year	Includes personnel, utilities, and maintenance.

LEAD TIME CALCULATION MODEL

Phase	Current (Before)	New Model (After)	Notes
Ocean Freight (VN → EU)	38 - 52 days	32 - 40 days	Route remains the same.
Customs Clearance at EU Port	2 - 3 days	2 - 3 days	Unchanged.
Last-mile Delivery (Port → Customer)	5 - 7 days	1 - 2 days	Greatest improvement as stock is already available at the Hamburg/Genoa hubs.
Total Lead Time (From customer order)	45 - 62 days	~3 - 5 days (if stock available)	~30-35 days (replenishment lead time)